

Epicenter Retreat 2012

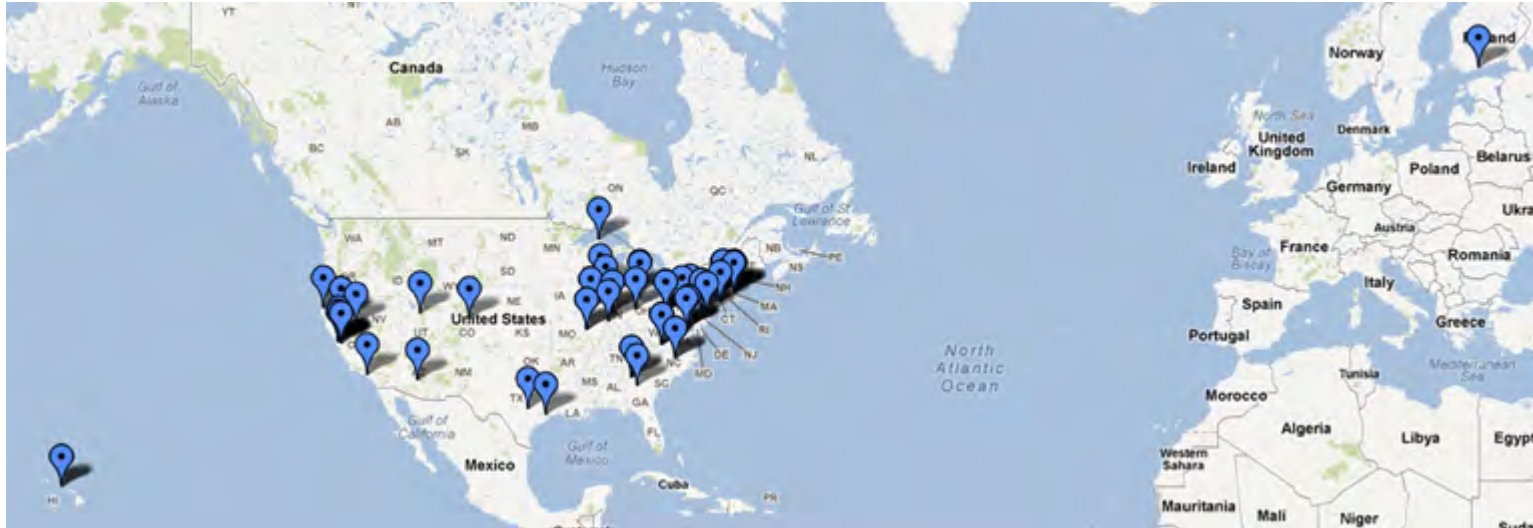
*Stanford Sierra Camp
Fallen Leaf Lake, Calif.
September 30 - October 3*



Hosted at the Stanford Sierra Camp near Lake Tahoe, this retreat was an opportunity for entrepreneurship and engineering faculty and industry experts to come together for spontaneous inspiration and interactions around creating a nation of entrepreneurial engineers.

The retreat utilized an “unconference” format, designed for active peer-to-peer collaboration. Participants had the opportunity to pitch topics of their interest, engage in small-group discussions on the topics proposed, design new curricular and extracurricular activities, and develop alliances and networks.

Retreat Participants



PUBLIC SCHOOL	SKC MGN LK	JRG	SA MB CH EC LG TA JG	KB KH IQ EH	JB WCN GB	GL
	MBS TB GT JD BP JBO LBC	SS LN SC RR	CB JP JT WK DC KT FH EF	NF NR RD	PJW OT ES DS MK CC	
PRIVATE SCHOOL	INDUSTRIAL & MECHANICAL ENGINEERING & DESIGN	OTHER ENGINEERING	BUSINESS SCHOOL	OTHER		

- 73 participants (27 women, 46 men)
- 36 schools, 4 NGOs, 1 governmental agency, 1 company
- faculty teaching over 8 engineering disciplines, as well as entrepreneurship, design, management and innovation



Stephanie Adams
Virginia Tech



Bridget Akinc
MIT



Beth Altringer
Harvard University



Tomasz Arciszewski
George Mason University



Anuradha Basu
San Jose State University



Gail Bentkover
U.S. EPA



Mary Besterfield-Sacre
University of Pittsburgh



Cheryl Bodnar
University of Pittsburgh



Marshall Brain
North Carolina State University



Leticia Britos Cavagnaro
Epicenter



Jason Bronowitz
Arizona State University



Tom Byers
Epicenter



Don Carpenter
Lawrence Technological University



Ed Carryer
Stanford University



Song Choi
University of Hawaii at Manoa



Sridhar Condoor
Saint Louis University



Emanuel Costache
Sagefox, Epicenter



Richard Cox
Epicenter



Edward Coyle
Georgia Tech



Richard Donnelly
George Washington University



Nathalie Duval-Couetil
Purdue University



Rebecca Edwards
Epicenter



Kathleen Eisenhardt
Epicenter



Daniel Ferguson
Purdue University



Eden Fisher
Carnegie Mellon University



Nathan Furr
Brigham Young University



Paula Gangopadhyay
The Henry Ford



John Gershenson
Michigan Technological University



Mehrdad Ghasemi-Nejhad
University of Hawaii at Manoa



Forrest Glick
Epicenter



Joy Godesiabo
Colorado School of Mines



Lonny Grafman
Humbolt State University



Fawwaz Habbal
Harvard University



Kazumasa Haneda
Aalto University



Cory Hixson
Virginia Tech



Cornelia Huellstrunk
Princeton University



Elizabeth Kisenwether
Penn State University



William Kline
Rose-Hulman
Institute of Technology



Mallory Kolinski
Sierra Nevada College



Greg Lank
U.S. EPA



John Lesko
Virginia Tech



Doug Melton
Kern Family Foundation



Laurie Moore
Epicenter



Holly Mott
NCIIA, Epicenter



Lawrence Neeley
Olin College



Wendy Newstetter
Georgia Tech



Steven Nichols
University of Texas, Austin



David Niño
Rice University



John Ochs
Lehigh University



Jack Patzer
University of Pittsburgh



Bob Podlasek
Bradley University



Iris Quan
San Jose State University



**Ramachandran
Radharamanan**
Mercer University



Nik Rokop
Illinois Institute of Technology



Jon Rust
North Carolina State University



Nikkie Salgado
Epicenter



Tina Seelig
Epicenter



Davor Senkovic
Epicenter



Angela Shartrand
NCIIA, Epicenter



Eli Shell
Epicenter



Sheri Sheppard
Epicenter



Steven Shooter
Bucknell University



Ikhlq Sidhu
UC Berkeley



Kurt Thoroughman
Washington University in St. Louis



Ognjen Todic
Epicenter



Greg Tonkay
Lehigh University



Andrew Tonsor
Caterpillar Inc.



Joe Tranquillo
Bucknell University



Olga Trusova
Epicenter



David Vorp
University of Pittsburgh



Phil Weilerstein
NCIIA, Epicenter



John-David Yoder
Ohio Northern University



Elaine Hagan
UCLA

Epicenter Retreat: preliminary survey

These data synthesize responses provided by 39 of the participants in a survey completed **in preparation** for the Retreat.



About the Respondents

Faculty or Administrator	72% Faculty 21% Admin
Years since highest degree	77% 11 or more
ASEE members	51% ASEE members
ASEE entrepreneurship & innovation division (ENT) members	53%
Recent entrepreneurial endeavor	76% within last three years
Who are your students?	77% teach undergraduates 46% teach graduates 23% include as part of course(s)
How do you teach entrepreneurship and innovation (E&I)?	57% course(s) focused on E&I
Ability to comment about E&I at your institution	88% very well 13% from a limited perspective



Definition of Entrepreneurship

Which of the following do you associate with **entrepreneurial engineers**?

Venture Creation, **Risk Tolerance**, Economic Development, Need for autonomy, Decision-making, Self-efficacy, **Opportunity Recognition**, Social orientation, Endurance, Market awareness, Customer development, Problem-solving, Ambiguity tolerance, **Creativity**, Influence and persuasion, **Product Development**, Startup finance and accounting, Oral/written communication, Teamwork, Leadership, **Value Creation**

(the items in bold represent the top 5 choices by the group of respondents; order is random)



Entrepreneurship & Engineering

Important measures for advancing E&I education for engineers % important & very important	71% teaching educators how to teach E&I 96% funding for E&I education 93% policies to encourage E&I education
E&I's role for undergrads % agreeing	75% required for undergrads 96% available as elective courses 89% available as extracurriculars
Challenges to making E&I a core part of undergraduate engineering education	• Lack of room in curriculum • Disagreement over E&I's value • Lack of funding



Needs & Resources

Which of the following resources would be most useful to you? (☑: top 5 selected by the group)

- ☒ Tools for assessing entrepreneurship and innovation learning
- ☒ Training in new entrepreneurship and innovation pedagogy
- ☒ Content to be integrated into existing lessons and courses
- ☒ Models for developing entrepreneurship and innovation initiatives
- ☒ Membership in a community of entrepreneurship and innovation thinkers and practitioners
- ☐ Content for creating new lessons and courses
- ☐ Content for extracurricular activities
- ☐ High-quality, Epicenter-endorsed online courses on entrepreneurship and innovation available for you and your students
- ☐ Research-based evidence of the value and impact of entrepreneurship and innovation in engineering
- ☐ Regional and national events, workshops, and competitions
- ☐ Curriculum for entire courses in entrepreneurship and innovation
- ☐ Programs for university leadership on integrating entrepreneurship and innovation into the curriculum



About the Respondents' Institutions

Type of institution	51% private / 49% public 80% award doctorates
Student engagement in E&I % frequently & very frequently	13% via core course 56% via elective course 50% via extracurricular activities
Administrative support for E&I	60% supportive
Institutional plan for E&I in Engineering	63% have a plan 30% actively implementing
Faculty teaching E&I	50% high engagement 19% significant number
Rate your Department's current practices % good & very good	23% teaching engineering educators how to teach E&I 30% funding for E&I education 43% policies encourage E&I education

Discussion Groups

Participants pitched 48 discussion topics to the group, related to infusing entrepreneurship and innovation into undergraduate engineering education. Related topics were pooled together to create 19 discussion groups. Topics comprised pedagogical approaches, programmatic models, theoretical frameworks, co-curricular initiatives and educational resources.

Participants moved fluidly between discussion groups, following “the law of the two feet,” by which participants were encouraged to use their two feet and find a place where they could contribute and add the most value.

Discussion Group 1:

- Faculty development (expand the team)
- Is experience needed to teach entrepreneurship?

Discussion Group 2:

- Experiential Innovation Modules
- Framing Senior Design projects as Innovation projects
- Experience Shape Mindset
- Innovation through Problem Based Learning

Discussion Group 3:

- Industry/academic partnerships
- Internships of a type your parents never had

Discussion Group 4:

- Gamification of instruction
- Student participation in classroom

Discussion Group 5:

- Reducing credit hours to add new cool things
- How does all of this fit into the curriculum?

Discussion Group 6:

- Dealing with resistant administrations
- How do we work across disciplines from an administrative perspective?

Discussion Group 7:

- Entrepreneurship versus innovation
- Business skills for engineers
- Design + entrepreneurship
- Management science of innovation

Discussion Group 8:

- Engineering innovativeness

Discussion Group 9:

- Empowering Students

Discussion Group 10:

- Foundation Course
- First year student issues and perceptions
- Teaching innovation in first year experience
- Hierarchical training from high school internship to undergraduate development

Discussion Group 11:

- Shared vision, shared action, shared impact
- Strategies for change in engineering education
- Vision for change

Discussion Group 12:

- Meta framework for innovation + entrepreneurship education
- Identification and assessment of skills associated with entrepreneurship thinking and problem solving (measurement)
- What are the core things that all students should know about entrepreneurship
- Fostering discussion of role of entrepreneurship (technology, social, culture)

Discussion Group 13:

- Creative Intelligence
- Innovation Canvas (develop concepts and sketch)

Discussion Group 14:

- Deep understanding of customer needs
- Revolutionize innovation commercialization
- Supporting student entrepreneurs

Discussion Group 15:

- Create prototype first

Discussion Group 16:

- Co-curricular: support and get out of the way
- Productive engineering entrepreneurship competitions
- Three day start-up
- Summer enrichment camps on innovation for undergrads
- Coaching engineering students to think creatively

Discussion Group 17:

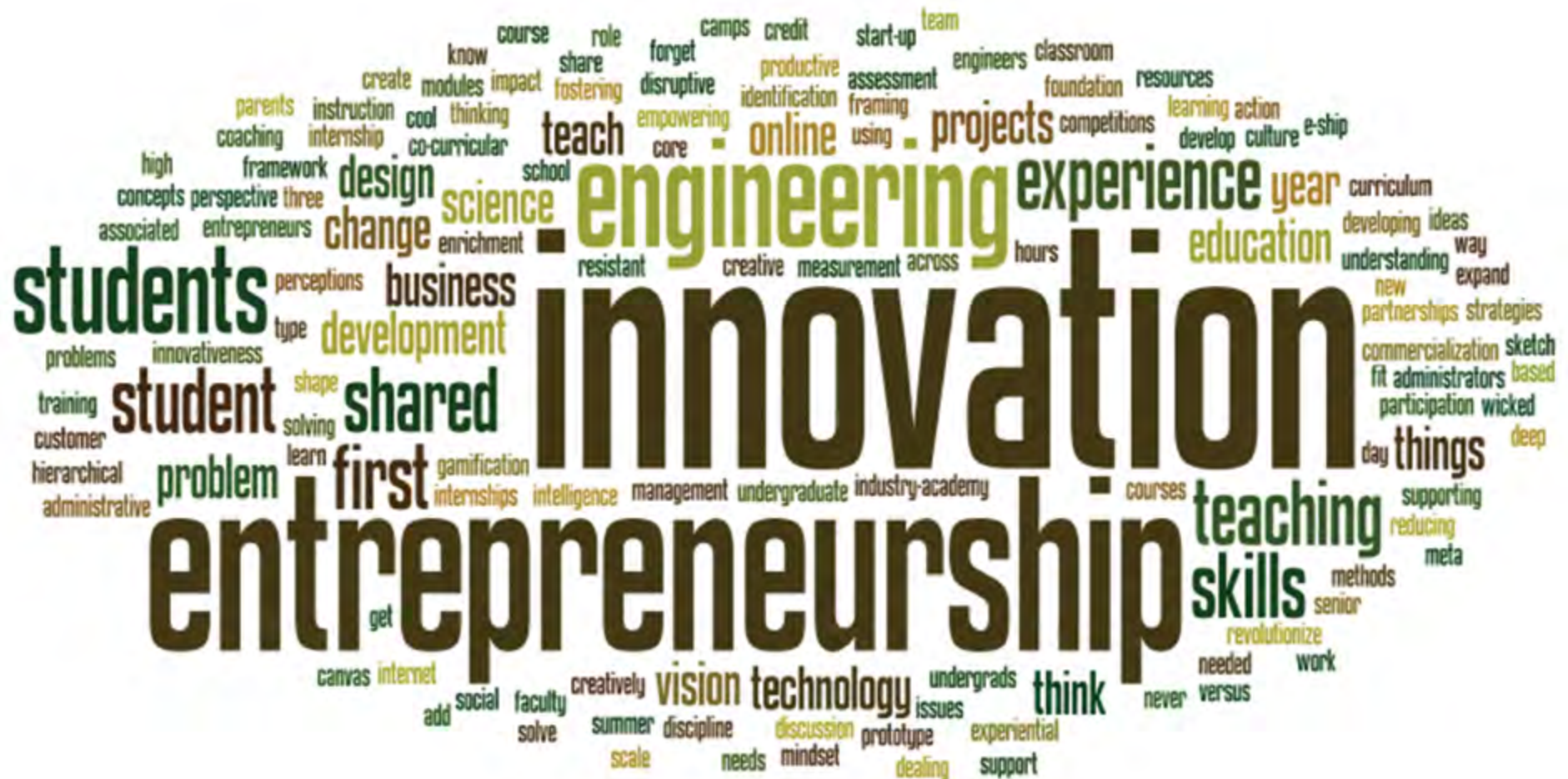
- Teach/learn how to think/solve wicked problems
- Developing disruptive technology (innovation: business + engineering)

Discussion Group 18:

- Online courses
- Teaching entrepreneurship skills online
- Using internet to share/scale teaching methods, ideas and resources

Discussion Group 19:

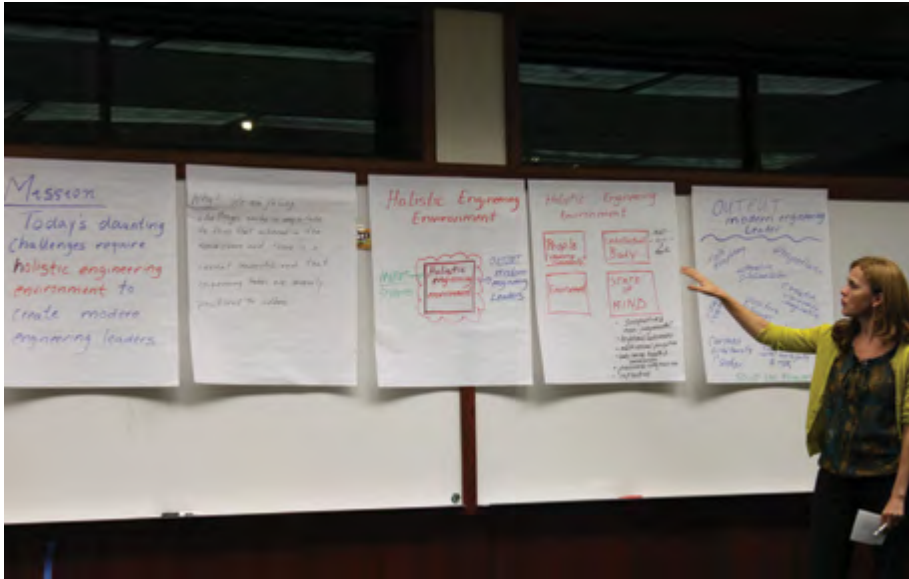
- Don't forget science students



This cloud represents the words used most frequently in discussion group topics

Working Groups

Based on the unconference exchanges, 10 working teams were formed. After three (or more) hours of work ideating and building prototypes, the teams presented summaries of their projects at the end of day 3. After the presentations, participants signed up to continue working on projects of their interest.



1. Creating Holistic Leaders

Description:

Today's daunting challenges require a holistic engineering environment to create modern engineering leaders

Audience:

Initially, global academic leaders initiating idea generation and process to finally benefit all of society

Team:

Tomasz Arciszewski, Andy Tonsor, Cornelia Huellstrunk, Mallory Kolinski, Fawwaz Habbal



2. Entrepreneurship Tool for ABET (ETA)

(Enhancing your case for accreditation)

Description:

Exploring how the entrepreneurial mindset may be mapped to ABET criteria

Audience:

Faculty, curriculum bodies, ABET

Team:

Liz Kisenwether, Greg Tonkay, Tom Byers, Stephanie Adams, Nathalie Duval-Coetil, Joe Tranquillo, Elaine Hagan



3. Creating an “Innovation Canvas”

Description:

Canvas that considers the key aspects of the innovation process, built off the Business Model Canvas (BMC) - we emphasize not having an order.

Audience:

Faculty/instructors

Team:

Bill Kline, Cory Hixson



4. Commercialization

Description:

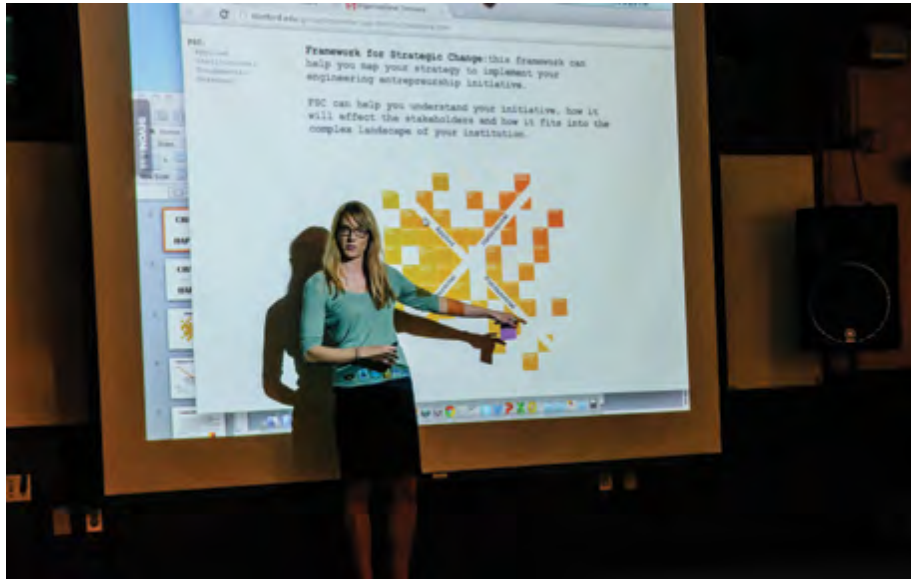
Student teams have a tough time taking their ideas to market. We use Business Model Canvas (BMC) as a core concept to develop a support system.

Audience:

Students with high potential ideas, serious about commercialization

Team:

Nik Rokop, Nathan Furr, John Ochs, Song Choi, Mehrdad Nejhad, Phil Weilerstein, Ed Coyle



5. How Change Actually Happens in Universities

Description:

Framework for entrepreneurship change (getting it through institutions and individual barriers)

Team:

Beth Altringer, Jack Lesko, Nikkie Salgado, Olga Trusova



6. The Conceptual Basis of Entrepreneurship Education - D³

Description:

Mapping the conceptual terrain of entrepreneurship in engineering education

Audience:

Engineering entrepreneurship faculty and students

Team:

Wendy Newstetter, Ikhlal Sidhu, Angela Shartrand, Dick Donnelly, Dan Ferguson, Eden Fisher, Paula Gangopadhyay, Kazumasa Haneda, David Niño, Marshall Brain



7. Online Entrepreneurship Course Builder

Description:

An engine to aggregate online entrepreneurship content

Audience:

- Faculty building their own courses
- Individual consumers/content generators

Team:

Doug Melton, Jason Bronowitz, Anuradha Basu, Ogi Todic, Forrest Glick, Tina Seelig



8. The Perfect Introduction to Engineering Entrepreneurship Module

Description:

An interview-based, web-supported student and facilitator module design tool

Audience:

Facilitators and students

Team:

Lawrence Neeley, JD Yoder, Mary Besterfield-Sacre, Rebecca Edwards, Bridget Akinc



9. Student AdVenture Camp

Description:

A scalable, immersive experience for empowering the next generation of innovators

Audience:

Students

Team:

Kurt Thoroughman, Gail Bentkover, Jack Patzer, Don Carpenter, John Gersheson, Laurie Moore



10. Customer Values in a World Without Oil

Description:

Utilizing a game-based technique we are teaching students how to speak with customers to determine their needs and value proposition.

Audience:

Students (multidisciplinary)

Team:

Cheryl Bodnar, Iris Quan, Steve Shooter, Joy Godesiabo, Sri Condoor, Ramachandran Radharamanan

About Epicenter

The National Center for Engineering Pathways to Innovation (Epicenter) is an education, research and outreach hub for the creation and sharing of entrepreneurship and innovation resources among engineering schools in the United States.

Funded by the National Science Foundation and directed by the Stanford Technology Ventures Program, Epicenter is dedicated to preparing undergraduate engineers to be more entrepreneurial and become bold innovators with the knowledge, skills and attitudes to contribute to the prosperity of the U.S. economy and society.

Learn more at *epicenter.stanford.edu*.



twitter.com/epicenterUSA



facebook.com/epicenterUSA



flickr.com/epicenterUSA

*Retreat photo credits: Cindy Pearson, Joe Tranquillo,
Holly Mott, Eli Shell, Laurie Moore, Olga Trusova*

Epicenter Team

Tom Byers, Principal Investigator

Kathleen Eisenhardt, Principal Investigator

Sheri Sheppard, Principal Investigator

Tina Seelig, Director

Leticia Britos Cavagnaro, Associate Director

Forrest Glick, Chief Information Architect

Laurie Moore, Communications Specialist

Olga Trusova, User Experience Design Lead

Ognjen Todric, Website Developer

Davor Senkovic, Programmer

Eli Shell, Digital Storyteller

Phil Weilerstein, NCIIA Executive Director

Angela Shartrand, NCIIA/Epicenter Research & Assessment

Alan Peterfreund, SageFox (External Evaluator)

Emanuel Costache, SageFox (External Evaluator)